# Curriculum Vitae

## Personal Data


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| Name: | Philine Gabriele Doris Feulner |
| Affiliations: | Eawag<br>[1]Swiss Federal Institute of Aquatic Science and Technology<br>Center for Ecology, Evolution and Biogeochemistry<br>Department of Fish Ecology & Evolution<br>Seestrasse 79, 6047 Kastanienbaum, Switzerland<br>[2]University of Bern<br>Institute of Ecology & Evolution<br>Baltzerstrasse 6, 3012 Bern, Switzerland |
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## Education

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| 07/2006 | Dissertation in Evolutionary Biology |
| 2003 - 2006 | PhD at the University of Potsdam |
| 11/2002 | Diploma in Biology |
| 1996 - 2002 | Graduate studies at the Christian-Albrechts-University, Kiel |

## Employment

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| Since 08/2014 | Group leader (tenure track) Fish Genomics at Eawag, Swiss Federal Institute of Aquatic Science and Technology |
| 01/2013 – 07/2014 | Postdoctoral researcher at the Max Planck Institute for Evolutionary Biology, Department Evolutionary Ecology, Plön |
| 01/2010 – 12/2012 | Postdoctoral researcher at the Westfälische Wilhelms University, Münster |
| 12/2007 - 11/2009 | Postdoctoral researcher on a Marie Curie Transfer of Knowledge host fellowship (PI Jon Slate) at University of Sheffield |
| 06/2006 - 11/2007 | Postdoctoral researcher at University of Potsdam |
| 01/2003 - 05/2006 | PhD student at University of Potsdam |

## Grants

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| Since 09/2015 | PI on SNSF grant: "Speciation Genomics of the Swiss Alpine Whitefish Radiation" (co-PI Ole Seehausen) |
| Since 08/2015 | PI on LEAD agency grant (evaluated within the German Science Foundation Priority Program 1819): "Host virus coevolution - demography versus selection" (co-PI Lutz Becks, MPI Plön) |
| 03/2011 to 02/2013 | PI on Marie Curie European Re-integration Grant (MC-ERG; call: FP7-PEOPLE-2010-RG; proposal n° 270891): "Structural Variation and Adaptation of the Stickleback Genome" |

## *List of publications*

Google Scholar profile:
https://scholar.google.ch/citations?user=fjfDDI8AAAAJ&hl=de&oi=ao
Citations 1439
h-index 16

1. Peer-reviewed articles

Frickel J., **Feulner P.G.D.**, Karakoc E., Becks L. (2018)
Population size changes and selection drive patterns of parallel evolution in a host–virus system.
*Nature Communications* 1706. doi:10.1038/s41467-018-03990-7

De-Kayne R., **Feulner P.G.D.** (2018)
A European whitefish linkage map and its implications for understanding genome-wide synteny between salmonids following whole genome duplication
bioRxiv 310136; doi: https://doi.org/10.1101/310136

**Feulner P.G.D.** *, Julia Schwarzer J. *, Haesler M.P., Meier J.I., Seehausen O. (2018)
A dense linkage map of Lake Victoria cichlids improved the Pundamilia genome assembly and revealed a major QTL for sex-determination
Accepted at G3, bioRxiv 275388; doi: https://doi.org/10.1101/275388

Matthews B., Best R.J., **Feulner P.G.D.**, Narwani A., Limberger R. (2017)
Evolution as an ecosystem process: insights from genomics.
*Genome* 61: 298-309

Stapley J., **Feulner P.G.D.**, Johnston S.E., Santure A.W., Smadja C.M. (2017)
Variation in recombination frequency and distribution across eukaryotes: patterns and processes.
*Philosophical Transactions of the Royal Society B* 20160455.

**Feulner P.G.D**., De-Kayne R. (2017)
COMMENTARY: Genome evolution, structural rearrangements and speciation.
*Journal of Evolutionary Biology* 30: 1488-1490.

Huang Y., Chain F.J.J., Panchal M., Eizaguirre C., Kalbe M., Lenz T.L., Samonte I.E., Stoll M., Bornberg-Bauer E., Reusch T.B.H., Milinski M., **Feulner P.G.D.** (2016)
Transcriptome profiling of immune tissues reveals habitat-specific gene expression between lake and river sticklebacks.
*Molecular Ecology* 25: 943–958

Lamanna F., Kirschbaum F., Ernst A.R.R., **Feulner P.G.D.**, Mamonekene V., Paul C., Tiedemann R. (2016)
Species delimitation and phylogenetic relationships in a genus of African weakly-electric fishes (Osteoglossiformes, Mormyridae, *Campylomormyrus*).
*Molecular Phylogenetic and Evolution* 101: 8-18.

Paul C., Mamonekene V., Vater M., **Feulner P.G.D.**, Engelmann J., Tiedemann R., Kirschbaum F. (2015)

Comparative histology of the adult electric organ among four species of the genus *Campylomormyrus* (Teleostei: Mormyridae).
*Journal of Comparative Physiology A* 201: 357–374.

**Feulner P.G.D.***, Chain F.J.J.*, Panchal M. *, Huang Y., Eizaguirre C., Kalbe M., Lenz T.L., Samonte I.E., Stoll M., Bornberg-Bauer E., Reusch T.B.H., Milinski M. (2015)
Genomics of divergence along a continuum of parapatric population differentiation.
*PLoS Genetics* 11: e1004966.

Chain F.J.J. *, **Feulner P.G.D**. *, Panchal M.*, Eizaguirre C., Samonte I.E., Kalbe M., Lenz T.L., Stoll M., Bornberg-Bauer E., Milinski M., Reusch T.B.H. (2014)
Extensive copy-number variation of young genes across stickleback populations.
*PLoS Genetics* 10: e1004830.

**Feulner P.G.D.***, Chain F.J.J.*, Panchal M.*, Eizaguirre C., Kalbe M., Lenz T.L., Mundry M., Samonte I.E., Stoll M., Milinski M., Reusch T.B.H., Bornberg-Bauer E. (2013)
Genome-wide patterns of standing genetic variation in a natural marine population of three-spined sticklebacks.
*Molecular Ecology* 22: 635-49.

**Feulner P.G.D. ***, Gratten J.*, Kijas J.W., Visscher P.M., Pemberton J.M., Slate J. (2013)
Introgression and the fate of domesticated genes in a wild mammal population.
*Molecular Ecology* 22: 4210-4221.

Schroeder J., Dugdale H.L., Radersma R., Hinsch M., Buehler D.M., Saul J., Porter L., Liker A., De Cauwer I., Johnson P.J., Santure A.W., Griffin A.S., Bolund E., Ross L., Webb T.J., **Feulner P.G.D.**, Winney I., Szulkin M., Komdeur J., Versteegh M.A., Hemelrijk C.K., Svensson E.I., Edwards H., Karlsson M., West S.A., Barrett E.L.B., Richardson D.S., van den Brink V., Wimpenny J.H., Ellwood S.A., Rees M., Matson K.D., Charmantier A., dos Remedios N., Schneider N.A., Teplitsky C., Laurance W.F., Butlin R.K., Horrocks N.P.C. (2013)
Fewer invited talks by women in evolutionary biology symposia.
*Journal of Evolutionary Biology* 26: 2063-2069.

Mundry M., Bornberg-Bauer E., Sammeth M., **Feulner P.G.D**. (2012)
Evaluating characteristics of *de novo* assembly software on 454 transcriptome data: a simulation approach.
*PLoS ONE* 7: e31410.

Stapley J.*, Reger J.*, **Feulner P.G.D**., Smadja C., Galindo J., Ekblom R., Bennison C., Ball A.D., Beckerman A., Slate J. (2010)
Adaptation Genomics: the next generation.
*Trends in Ecology and Evolution* 25: 705-712.

Slate J., Santure A.W., **Feulner P.G.D**., Brown E.A., Ball A.D., Johnston S.E., Gratten J. (2010)
Genome mapping in intensively studied wild vertebrate populations.
*Trends in Genetics* 26: 275-284.

**Feulner P.G.D.**, Plath M., Engelmann J., Kirschbaum F., Tiedemann R. (2009)
Electrifying love: electric fish use species-specific discharge for mate recognition.
*Biology Letters* 5: 225-228.

Skog A., Zachos F.E., Rueness E.K., **Feulner P.G.D.**, Mysterud A., Langvatn R., Lorenzini R., Hmwe S.S., Lehoczky I., Hartl G.B., Stenseth N.C., Jakobsen K.S. (2009) Phylogeography of red deer (*Cervus elaphus*) in Europe.
*Journal of Biogeography* 36: 66-77.

**Feulner P.G.D.**, Kirschbaum F., Tiedemann R. (2008)
Adaptive radiation in the Congo River: An ecological speciation scenario for African weakly electric fish (Teleostei; Mormyridae; *Campylomormyrus*).
*Journal of Physiology – Paris* 102: 340-346.

Tobler M., DeWitt T.J., Schlupp I., García de León F.J., Hermann R., **Feulner P.G.D.**, Tiedemann R., Plath M. (2008)
Toxic hydrogen sulfide and dark caves: phenotypic and genetic divergence across two abiotic environmental gradients in *Poecilia mexicana*.
*Evolution* 62: 2643-2659.

**Feulner P.G.D.**, Kirschbaum F., Mamonekene V., Ketmaier V., Tiedemann R. (2007)
Adaptive radiation in African weakly electric fish (Teleostei: Mormyridae: *Campylomormyrus*): a combined molecular and morphological approach.
*Journal of Evolutionary Biology* 20: 403-414.

**Feulner P.G.D.**, Kirschbaum F., Schugardt C., Ketmaier V., Tiedemann R. (2006)
Electrophysiological and molecular genetic evidence for sympatrically occuring cryptic species in African weakly electric fishes (Teleostei: Mormyridae: *Campylomormyrus*).
*Molecular Phylogenetics and Evolution* 39: 198-208.

**Feulner P.G.D.**, Kirschbaum F., Tiedemann R. (2005)
Eighteen microsatellite loci for endemic African weakly electric fish (*Campylomormyrus*, Mormyridae) and their cross species applicability among related taxa.
*Molecular Ecology Notes* 5: 446-448.

**Feulner,P.G.D.**, Bielfeldt W., Zachos F.E., Bradvarovic J., Eckert I., Hartl G.B. (2004)
Mitochondrial DNA and microsatellite analyses of the genetic status of the presumed subspecies *Cervus elaphus montanus* (Carpathian red deer).
*Heredity* 93: 299-306.

2. Other relevant publications

Stapley J., **Feulner P.G.D.**, Johnston S.E., Santure A.W., Smadja C.M. (2017)
Recombination: the good, the bad and the variable.
*Philosophical Transactions of the Royal Society B* 20170279.

Chain F.J.J., **Feulner P.G.D.** (2014)
Ecological and evolutionary implications of genomic structural variations.
*Frontiers in Genetics* 5: 326.

Tiedemann R., **Feulner P.G.D**., Kirschbaum F. (2010)
Electric organ discharge divergence promotes ecological speciation in sympatrically occurring African weakly electric fish (*Campylomormyrus*).
*Evolution in Action, Springer Verlag* (p 307-321)

**Feulner P.G.D.**, Plath M., Engelmann J., Kirschbaum F., Tiedemann R. (2009) Article Addendum - Magic trait Electric Organ Discharge (EOD): Dual function of electric signals promotes speciation in African weakly electric fish.
*Communicative & Integrative Biology* 2: issue 4.

* authors contributed equally